

Rishi Vanukuru

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University of Colorado Boulder

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RESEARCH SUMMARY

As a designer and human-computer interaction researcher, I build and study new interactive systems that support meaningful collaboration across space and over time. With new technologies seeking to radically change the way we live and work, I believe we have a responsibility to take a more critical and intentional stance towards design. Drawing on insights from the fields of engineering, design, and the humanities, I create tools and tool systems for use in the immediate future, and demonstrate possible alternatives to the dominant implementations of technology today.

EDUCATION

- 2022–26 **University of Colorado Boulder**
Ph.D. in Creative Technology & Design
- 2018–20 **Indian Institute of Technology Bombay**
M.Des. in Interaction Design
- 2014–18 **Indian Institute of Technology Bombay**
B.Tech. in Civil Engineering
Minor in Design

RESEARCH EXPERIENCE

- ATLAS Institute, University of Colorado Boulder** 2022 – 2026
Graduate Research Assistant
Advisor: Dr. Ellen Yi-Luen Do
Dissertation Title: *Designing the Everyday Spatial Computer: Supporting Meaningful Collaboration using Mobile Devices* [Thesis]
- Microsoft Research, Cambridge, UK** Summer 2024
Research Intern
Advisor: Dr. Sean Rintel
Designing Interfaces that Support Temporal Work Across Meetings with AI
- Ericsson Research, Santa Clara, USA** Summer 2023
Research Intern
Advisor: Dr. Gregoire Phillips
Systems for Network-adaptive Augmented Reality Remote Communication
- IDC School of Design, IIT Bombay** Fall 2020 – Spring 2021
Research Associate
Advisor: Dr. Jayesh Pillai
Supporting Remote Learning with Mobile Augmented Reality
- Laval Virtual Institute, Arts et Métiers, France** Summer 2019
Research Intern
Advisor: Dr. Simon Richir
Studying Creativity and Design in Virtual Reality

SELECTED PUBLICATIONS

Journal & Conference Articles

- 2026 **Rishi Vanukuru**, Krithik Ranjan, Ada Yi Zhao, David Lindero, Gunilla Berndtsson, Gregoire Phillips, Amy Banic, Mark D. Gross, and Ellen Yi-Luen Do. “Studying Mobile Spatial Collaboration across Video Calls and Augmented Reality”. In: *Proceedings of the ACM on Human-Computer Interaction*. 10, 2, Article CSCW037 (May 2026). doi:10.1145/3788073
- 2026 Christoph Anthes, Mark Billingham, Uwe Grünefeld, Hans-Christian Jetter, Hai-Ning Liang, Frank Maurer, David Aigner, Craig Anslow, Guillaume Bataille, Abraham G. Campbell, Judith Friedl-Knirsch, Alexander Gall, Renan Guarese, Sebastian Hubenschmid, Yue Li, Fabian Pointecker, Andreas Riegler, Daniel Roth, **Rishi Vanukuru**, Nanjia Wang, Lingyun Yu, Johannes Zagermann, and Daniel Zielasko. “Grand Challenges in Cross Reality”. In: *IEEE Transactions on Visualization and Computer Graphics (Early Access)*. doi:10.1109/TVCG.2026.3695303
- 2025 **Rishi Vanukuru**, Payod Panda, Xinyue Chen, Ava Elizabeth Scott, Lev Tankelevitch, and Sean Rintel. “Designing Interfaces that Support Temporal Work Across Meetings with Generative AI.” In: *Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS ’25)*. doi:10.1145/3715336.3735833.
- 2025 Xinyue Chen, Lev Tankelevitch, **Rishi Vanukuru**, Ava Elizabeth Scott, Payod Panda, and Sean Rintel. “Are We On Track? AI-Assisted Active and Passive Goal Reflection During Meetings.” In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI ’25)*. doi:10.1145/3706598.3714052
- 2025 Ava Elizabeth Scott, Lev Tankelevitch, Payod Panda, **Rishi Vanukuru**, Xinyue Chen, and Sean Rintel. “What Does Success Look Like? Catalyzing Meeting Intentionality with AI-Assisted Prospective Reflection.” In: *Proceedings of the 4th Annual Symposium on Human-Computer Interaction for Work (CHIWORK ’25)*. doi:10.1145/3729176.3729204
- 2025 Krithik Ranjan, Anika Mahajan, **Rishi Vanukuru**, and Ellen Yi-Luen Do. “The Design Space of Tangible Interfaces for Computational Tinkerability.” In: *Constructionism Conference Proceedings*, 8, 325-340.. doi:10.21240/constr/2025/102.X
- 2023 **Rishi Vanukuru**, Suibi Che-Chuan Weng, Krithik Ranjan, Torin Hopkins, Amy Banic, Mark D. Gross, and Ellen Yi-Luen Do. “DualStream: Spatially Sharing Selves and Surroundings using Mobile Devices and Augmented Reality.” In: *IEEE International Symposium on Mixed and Augmented Reality, ISMAR 2023*. doi:10.1109/ISMAR59233.2023.00028
- 2022 S. Sandra Bae, **Rishi Vanukuru**, Ruhan Yang, Peter Gyory, Ran Zhou, Ellen Yi-Luen Do, and Danielle Albers Szafir. “Cultivating Visualization Literacy for Children Through Curiosity and Play.” In: *IEEE Transactions on Visualization and Computer Graphics*, 2022. doi:10.1109/TVCG.2022.3209442
- 2022 Torin Hopkins, Suibi Che-Chuan Weng, **Rishi Vanukuru**, Emma Wenzel, Amy Banic, Mark D. Gross, and Ellen Yi-Luen Do. “AR Drum Circle: Real-Time Collaborative Drumming in AR.” In: *Frontiers in Virtual Reality*, 91. doi:10.3389/frvir.2022.847284
- 2022 Torin Hopkins, Suibi Che-Chuan Weng, **Rishi Vanukuru**, Emma Wenzel, Amy Banic, and Ellen Yi-Luen Do. “Studying the Effects of Network Latency on Audio-Visual Perception During an AR Musical Task.” In: *IEEE International Symposium on Mixed and Augmented Reality, ISMAR 2022*. doi:10.1109/ISMAR55827.2022.00016
- 2021 Sylvain Fleury, **Rishi Vanukuru**, Charles Mille, Killian Poinot, Aurélien Agnès, and Simon Richir. “CRUX: A Creativity and User Experience model.” In: *Digital Creativity*, 32:2 (pp. 116-123). doi:10.1080/14626268.2021.1915339
- 2020 Sylvain Fleury, Aurélien Agnès, **Rishi Vanukuru**, Emma Goumillout, Nicolas Delcombel, and Simon Richir. “Studying the Effects of Visual Movement on Creativity.” In: *Thinking Skills and Creativity*, 36, 100661. doi:10.1016/j.tsc.2020.100661

Short Papers, Posters, & Demos

- 2024 **Rishi Vanukuru** and Ellen Yi-Luen Do. “Spatial Mobile Memories: Recording and Sharing Everyday Moments using Mobile Augmented Reality.” In: *Proceedings of the 2024 ACM International Conference on Interactive Media Experiences (IMX '24)*. doi:10.1145/3639701.3661087
- 2024 Torin Hopkins, Suibi Che-Chuan Weng, **Rishi Vanukuru**, Sasha Novack, Chad Tobin, Emma Wenzel, Amy Banic, Mark D. Gross, and Ellen Yi-Luen Do. “XR Jam: Design Considerations for Music Networking with AI in XR.” In: *2024 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR)*. doi:10.1109/AIxVR59861.2024.00021
- 2020 **Rishi Vanukuru**, Amarnath Murugan, and Jayesh Pillai. “Dual Phone AR: Using a Second Phone as a Controller for Mobile Augmented Reality.” In: *Adjunct Publication of the 33rd Annual ACM Symposium on User Interface Software and Technology (UIST '20 Adjunct)*. doi:10.1145/3379350.3416139
- 2020 **Rishi Vanukuru**. “Accessible Spatial Audio Interfaces: A Pilot Study into Screen Readers with Concurrent Speech.” In: *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)*. doi:10.1145/3334480.3381440

AWARDS

- 2022 Dr. Awtar and Teji Singh Graduate Fellowship (\$20,000)
- 2020 Institute Silver Medal, IIT Bombay (*1st rank in the M.Des program*)
- 2020 1st Position, CHI Student Research Competition (*Graduate category*)

TEACHING & SERVICE

Co-Instructor

Introduction to Virtual Reality (2022)

Teaching Assistant

Design for Immersive Media (2019-2021), Engineering Mechanics (2018)

Organizing Committee

Lead for Virtual Operations, IndiaHCI 2020

Reviewer

CHI 2026, 2025, 2023; DIS 2026, 2024; VIS 2026; VRST 2025; EuroVIS 2025; IEEE VR 2024; INTERACT 2023; CHI 2021 Student Research Competition

Student Volunteer

UIST 2020, IndiaHCI 2019, TypoDay 2019

Student Mentor

Indian Institute of Technology Bombay, 2016–2018

FUNDING PROPOSALS

Title: The Shared Reality Project: Towards an Ecosystem for Cross-device Holographic Communication

PI: Dr. Ellen Yi-Luen Do & Dr. Mark D. Gross

Funding agency: Ericsson Research

Funding amount: \$65,000

Title: Augmented Reality as an Interface for the Internet of Things and People

PI: Dr. Ellen Yi-Luen Do

Funding agency: Pervasive Personalized Intelligence Center, NSF IUCRC

Funding amount: \$56,788